

AI-Augmented Weatherwise API

ENTERPRISE METEOROLOGICAL INTELLIGENCE PLATFORM

Document Reference: WWISE-SPEC-2026-V4

Target Audience: Systems Engineers, ML Operators & API Integrators

1. Executive Summary & Platform Overview

The AI-Augmented Weatherwise API provides an enterprise-grade platform for real-time weather intelligence and hyper-local predictive analytics. By merging numerical weather prediction (NWP) physics with 3D spatial-temporal Transformer neural networks, the system delivers forecast latencies under 50ms per query across global targets with sub-kilometer accuracy.

2. Global System Architecture

The distributed cloud architecture spans multi-region availability zones designed for active-active redundancy and 99.999% SLA availability. Incoming telemetry streams from satellites, Doppler radar, buoy arrays, and terrestrial IoT stations ingest directly into high-throughput Apache Kafka clusters before TPU-based neural inference execution.

3. Spatial Gridding & Coordinates

All observational and predictive datasets standardize on the WGS 84 spatial reference system, gridded globally using Uber's H3 hexagonal spatial index at Resolution 9 (~0.1 sq km cell area). Vertical atmospheric profiles encompass 37 standard barometric pressure levels ranging from 1000 hPa surface to 10 hPa stratospheric limits.

Parameter	Specification	Target SLA / Performance
Global Uptime SLA	Multi-region Active-Active	99.999% Availability
Inference Latency	INT8 Quantized TPU Model	< 22ms Median Response
Spatial Grid Unit	H3 Hexagonal Indexing (Res 9)	0.1 km ² Cell Area
Data Ingestion Rate	Kafka Cluster Ingestion Pipeline	> 500,000 events/sec

4. Gateway Architecture & Security Protocols

Security is enforced at the network perimeter via Cloudflare Enterprise and an Envoy Proxy API Gateway layer. Mandatory mutual TLS (mTLS) or OAuth 2.0 bearer tokens are required on all routes. Rate limits operate on distributed sliding-window counters managed in Redis clusters.

- **Authentication:** OAuth 2.0 Bearer Tokens / HMAC-SHA256 Signed API Keys.
- **Transport Security:** Strict TLS 1.3 protocol requirement with cipher suite restrictions.
- **Rate Limiting:** Tiered token bucket enforcement (Standard: 1,000 req/min; Enterprise: Custom up to 500k req/min).
- **Protection:** Integrated WAF and DDoS mitigation capacity handling up to 2.5 Tbps.

5. Core Endpoint Specifications (REST & gRPC)

Endpoints support both standard OpenAPI 3.1 RESTful JSON interaction and high-throughput gRPC over HTTP/2 for ultra-low latency streaming applications.

Key API Routes:

- GET /v4/current: Instantaneous point weather observations with AI gap-filling.
- GET /v4/forecast/hourly: Hourly predictions up to 240 hours horizon.
- GET /v4/forecast/daily: Aggregated 45-day daily extended climate outlooks.
- GET /v4/nowcast/minutely: High-resolution 1-minute step forecast for immediate 120 minutes.
- GET /v4/map/tiles: XYZ vector and raster radar/satellite map overlays.

```
// HTTP Response Header Standard HTTP/1.1 200 OK Content-Type: application/json;
charset=utf-8 X-RateLimit-Limit: 10000 X-RateLimit-Remaining: 9984 X-RateLimit-Reset:
1774705935 ETag: W/"5f8a-92b81d7e"
```

6. Specialized Industry Intelligence Endpoints

Beyond standard meteorology, specialized modules cater directly to enterprise sector operations:

Aviation & Defense (/v4/aviation/sigmet)

Provides 3D polygon geometry volumes for flight hazards including clear air turbulence (CAT), icing levels, squall lines, and volcanic ash dispersion profiles aligned with ICAO Annex 3 standards.

Marine Operations (/v4/marine/ocean-state)

Tracks primary/secondary wave heights, swell periods, ocean currents, sea ice concentration, and thermocline depth layers for commercial maritime routing and offshore platform safety.

Agronomy & Hydrology (/v4/agronomy/soil-moisture)

Delivers volumetric water content across 4 subsurface depths (10cm to 200cm), soil temperatures, Evapotranspiration (ET0/ETc), and leaf wetness duration predictions for precision farming.

Renewable Energy (/v4/energy/solar-wind-power)

Converts forecasts directly into Megawatt-hour (MWh) power generation predictions for wind and solar farms, accounting for solar panel tilt, cloud shadow vectors, and wind turbine hub height speeds.

```
// Sample JSON Payload Snippet - Marine Endpoint { "location": { "lat": 34.0522, "lon":  
-118.2437 }, "ocean_state": { "significant_wave_height_m": 2.4, "primary_swell_period_s":  
11.2, "sea_surface_temp_c": 16.8 } }
```

7. Webhook Infrastructure & Real-time Alerting

Real-time event delivery bypasses polling mechanisms by delivering HMAC-SHA256 signed JSON notifications directly to subscriber HTTP endpoints upon severe weather triggers or threshold breaches.

```
// Webhook Payload Protocol { "event_id": "evt_99182a472c", "event_type":
"alert.severe_weather.warning", "timestamp": "2026-09-28T12:00:00Z", "data": {
"hazard_type": "Tornado Warning", "urgency": "Immediate", "severity": "Extreme",
"effective_until": "2026-09-28T13:30:00Z", "affected_polygon": [[-97.51, 35.46], [-97.35,
35.46], [-97.35, 35.60], [-97.51, 35.60]] } }
```

8. Error Standards & Troubleshooting

Errors follow RFC 7807 Problem Details standards. The matrix below outlines primary error condition responses:

Status Code	Error Subcode	Description & Resolution
400 Bad Request	invalid_parameters	Coordinates out of bounds or invalid date ranges.
401 Unauthorized	key_expired	API key missing or failed HMAC verification.
429 Rate Limited	quota_exceeded	Request bucket empty; wait for Retry-After period.
500 Internal Error	model_timeout	Neural inference failed; retry request with backoff.

9. SDK Libraries & Code Implementations

Official client SDKs simplify integration across languages by managing authentication, connection pooling, and retry mechanisms automatically.

Python Implementation Example

```
from weatherwise import WeatherClient, Units
client = WeatherClient(api_key="ww_live_987654321") # Fetch hourly forecast for location
forecast = client.forecast.get_hourly(latitude=37.7749, longitude=-122.4194, hours=24,
units=Units.METRIC)
for hour in forecast.hours:
    print(f"Time: {hour.time} | Temp: {hour.temp}°C | Rain Prob: {hour.pop}%")
```

Node.js / TypeScript Example

```
import { WeatherwiseClient } from '@weatherwise/sdk';
const client = new WeatherwiseClient({ apiKey: 'ww_live_987654321' });
async function getNowcast() {
    const nowcast = await client.nowcast.getMinutely({ lat: 40.7128, lon: -74.0060 });
    console.log(`Current dBZ: ${nowcast.minutes[0].reflectivity}`);
}
```

10. Operations, Observability & CI/CD

System deployment utilizes GitOps pipelines driven by ArgoCD and Kubernetes. Microservices maintain strict zero-downtime deployment capabilities via canary release stages monitored through OpenTelemetry metrics.

- **Observability:** Centralized ELK logging paired with Prometheus metrics and Grafana dashboards.
- **Tracing:** Distributed request correlation IDs trace calls end-to-end from edge proxies to TPU nodes.
- **Testing:** Continuous chaos engineering drills validate regional failover capabilities automatically.
- **Compliance:** Annual SOC 2 Type II audits, ISO 27001 certification, and strict GDPR/CCPA data privacy compliance.

11. Autonomous Navigation & Smart City Integration

Autonomous vehicles (AVs) and robotaxis leverage sub-kilometer surface friction metrics, hydroplaning risk indices, and optical fog/rainfall density values to adapt driving algorithms in real time.

Smart cities integrate API feeds to automate pavement de-icing salt applications, control storm water drainage pumps prior to heavy downpours, and optimize HVAC efficiency across municipal building networks.

12. SLA Terms, Pricing & Technical Support

Weatherwise offers flexible pricing tiers designed for startups through enterprise infrastructure deployments:

Plan Tier	Monthly Quota	Support Level	Uptime SLA
Developer Sandbox	100,000 requests	Community Forums	Best Effort
Professional	10,000,000 requests	Business Hours Email	99.9% Uptime
Enterprise Infrastructure	Custom / Unlimited	24/7 Phone + TAM	99.999% Guaranteed

13. Conclusion & Support Contact

The AI-Augmented Weatherwise API provides the ultimate foundation for meteorological intelligence, combining cutting-edge deep learning with modern API architecture.

Technical Support & Integration Inquiries:
Email: support@weatherwise.ai | Developer Portal: <https://developer.weatherwise.ai>

